Emine Senkardesler

emine2@illinois.edu

EDUCATION

PhD in Informatics, University of Illinois Urbana Champaign, 2029 (anticipated) **GPA:4.0**
MS in GGIT, Middle East Technical University, 2024
BS in City and Regional Planning, Middle East Technical University, 2019

EMPLOYMENT

University of Illinois Urbana Champaign Illinois, USA

- Teaching Assistant, Informatics Department, Crop Science Department January 2026 – Present
- Graduate Research Assistant, Informatics Department August 2024 - December 2025
  - RESILIENT AGRICULTURAL SUPPLY CHAINS AGAINST CLIMATE HAZARDS (RASCCH) – Analyzing crop insurance, yield and weather data to create mitigation scenarios for robust agricultural production.
  - PERMAFROST DISCOVERY GATEWAY – Identifying permafrost thaw risk areas for infrastructure and finding potential neighborhood extension places.

Farmlabs & Geosys 2019 – 2023
Remote Sensing and GIS Specialist *Ankara, TURKEY*

- Processed time series satellite data and created zone maps for variable rate fertilization.
- Conducted field work for ground truth and research improvement, communicated with stakeholders.
- Parcel Delineation
- Drought Analysis
- Project Proposals

Geosys 2018 – 2019
Research Intern *Ankara, Turkey*

- Supported research tasks in remote sensing and geospatial data analysis
- Building Urban Atlas

TECHNICAL STRENGTHS

Programming languages: Python, R
Tools and Frameworks: Geoserver, ArcGIS, QGIS, Fusion360, Figma
Databases: SQL
Others: Machine Learning, Deep Learning, Spatial Analysis, GeoAI

PUBLICATIONS

Not Peer-Reviewed or Abstracts in Conference Proceedings:

- "Wainwright Community – Permafrost Thaw Risk Assessment Project" IEEE International Conference on eScience (eScience) 2025
- "Estimating Corn Phenology by Integrating Object - Oriented Remote Sensing and Machine Learning to Create Field Model Environmental Smog", The 5th International Electronic Conference on Remote Sensing 2023

PRESENTATIONS

At Conferences:

- "Prediction of Crop Insurance Loss Ratio Caused by Extreme Moisture and Drought" 2026 AAG Annual Meeting, Presentation
- "Wainwright Community - Permafrost Thaw Risk Assessment Project", 2025 21st IEEE International Conference on e-Science, eScience 2025, Poster Presentation
- "Spatial and Temporal Analysis of Crop Frequency and Insurance Losses to Predict Crop Losses Under Certain Climate Conditions", 2025 AAG Annual Meeting, Presentation
- "Estimating Corn Phenology by Integrating Object-Oriented Remote Sensing and Machine Learning to Create Field Model", 2023 AAG West Lakes Division Meeting, Presentation
- "Estimating Corn Phenology by Integrating Object - Oriented Remote Sensing and Machine Learning to Create Field Model", The 5th International Electronic Conference on Remote Sensing 2023, Presentation

AWARDS AND HONORS

- Poster Award, 1st Place, Plant Science Symposium, 2026
- Applied Geography Specialty Group Annual Award, 2025
- Master's Paper Presentation 2nd Place, AAG West Lakes Division Meeting, 2023

TEACHING

Classes as TA:

- CPSC 442 Introduction to Programing with R, Fall 2026
- CPSC 444 Introduction to Spatial Analytics, Fall 2025, 2026
- CPSC 440 Applied Statistics 1, Spring 2026
- CPSC 499 Soft Skills, Spring 2026

PROFESSIONAL ORGANIZATIONS

METU Cycling Team, 2015-present